

Interview Questions — Java - Microservices with Spring Boot, Spring Cloud

Total Questions: 33

1. What is a monolithic application, and how does it differ from a microservices architecture?

Hint: Think about how all features are packaged and deployed in each approach.

2. List two main advantages of using microservices over a monolithic architecture.

Hint: Consider deployment and scalability.

3. Explain, in your own words, why independent deployment is important in a microservices setup.
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4. In the context of a campus management system, what could be two separate microservices you might create? Briefly explain their responsibilities.

Hint: Think about user-related tasks versus event management.

5. How would you create a simple REST API in Spring Boot for managing users (CRUD operations)? Briefly outline the main steps.

Hint: Mention project setup, controller, and repository.

6. What is the purpose of REST API communication between microservices?

Hint: Focus on how services interact and exchange data.

7. Write a basic Spring Boot controller method to handle a GET request for fetching all users.

Hint: Use `@GetMapping` in your answer.

8. What is Eureka in the Spring Cloud ecosystem, and why is it used?

Hint: Think about service registration and discovery.

9. Describe how a microservice registers itself with a Eureka server.

Hint: Focus on annotations and configuration.

10. How can one microservice discover the address of another microservice using Eureka?

Hint: Mention how clients fetch instance information programmatically.

11. If your UserService and EventService are both registered with Eureka, what happens when EventService goes down? How does Eureka help?

Hint: Think about service registry and availability.

12. What is the purpose of an API Gateway in a microservices architecture?

Hint: Consider how it manages requests between clients and services.

13. Explain how Spring Cloud Gateway routes a request to the appropriate microservice.

14. What is load balancing, and why is it important in microservices?

Hint: Think about distributing traffic among multiple service instances.

15. How does Ribbon work with Eureka to provide client-side load balancing?

Hint: Ribbon uses Eureka's service registry to choose an instance.

16. Suppose you want all requests to /events/** to go to the EventService. How would you configure this in Spring Cloud Gateway?

Hint: Think about route predicates and target URIs.

17. What is a Feign Client, and how does it simplify inter-service communication?

Hint: Focus on how it handles REST calls.

18. Write a simple Feign Client interface for calling the EventService's /events endpoint.

Hint: Use @FeignClient and method mapping.

19. How would you handle errors or failures when one microservice calls another using Feign?

Hint: Mention fallback methods or error handling strategies.

20. Give an example scenario where UserService needs to fetch event details from EventService. How would you implement this using Feign?

21. What is the purpose of Spring Boot Actuator in microservices?

Hint: Think about application monitoring and health checks.

22. How can Prometheus and Grafana be used together to monitor your microservices?

Hint: One collects metrics, the other visualizes them.

23. Which actuator endpoints would you enable to check the health and metrics of your microservices?

Hint: Think about `/actuator/health` and `/actuator/metrics`.

24. Describe how you would integrate Prometheus with a Spring Boot microservice for monitoring.

Hint: Consider dependencies and configuration.

25. In your mini project splitting Campus Connect into User and Event services, what role does Eureka play?

Hint: Think about how services find and communicate with each other.

26. How would you demonstrate API Gateway routing in your Campus Connect microservices project?

27. If your monitoring dashboard shows that EventService is unhealthy, what steps would you take to investigate and resolve the issue?

Hint: Consider logs, actuator endpoints, and service dependencies.

28. Suppose you need to add a new NotificationService to your Campus Connect microservices system. What steps would you follow to register it with Eureka and make it accessible via the API Gateway?

29. Integrative: How do API Gateway, Eureka, and Feign Client work together to enable seamless communication between UserService and EventService?

Hint: Think about routing, discovery, and REST calls.

30. Integrative: Explain how monitoring with Actuator and Prometheus can help you quickly identify and resolve issues in a load-balanced microservices environment.

Hint: Focus on metrics, health checks, and traffic distribution.

31. Integrative: In your Campus Connect project, how would you ensure that user-related requests are always routed to healthy instances of UserService, even if some instances go down?

Hint: Consider Eureka's registry and the load balancer's behavior.

32. How could you use an AI assistant like ChatGPT or GitHub Copilot to help you troubleshoot a failing microservice or debug a Spring Boot configuration issue?

Hint: Mention how you would verify AI suggestions before applying them.

33. If you use an AI tool to generate code for a Feign Client or an Actuator configuration, what would you check before adding it to your project?

Hint: Think about correctness, security, and compatibility.
